

Magnetic fields in matter

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Either due to angular momentum or spin $\rightarrow$ magnetic moment

$$\vec{M} = n \langle \vec{m} \rangle \quad (1)$$

$$\langle \vec{m} \rangle = \frac{1}{N} \sum_{i=1}^N \vec{m}(\vec{x}_i) = \frac{\text{av mag moment}}{\text{atom}} \quad (2)$$

Most materials have $\langle \vec{m} \rangle = 0$ if $\vec{B} = 0$ no external magnetic field.

If we apply a magnetic field then

$$\vec{B} \propto \vec{M} \quad (3)$$

$$\vec{M} = \frac{1}{\mu_0} \frac{\chi_m}{1 + \chi_m} \vec{B} \quad (4)$$

$$(5)$$

where χ_m is magnetic susceptibility.

Dag magnetic materials $-1 < \chi_m < 0$ e.g copper, gold, water. $\vec{M}$ aligned to $\vec{B}$ but opposite directions.

Paramagnetic materials $\chi_m > 0$ e.g aluminum and tungsten.

Ferromagnetic materials $\vec{M} \neq 0$ at $\vec{B} = 0$ e.g iron, nickel. (below curie temperature)

While $\chi_e > 0$, χ_m can have either sign.

Assume we have a magnetic moment then we will incorporate it into our Maxwell equations.

Magneto statics (Griffiths chapter 5)

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{x}}{r^3} \quad (6)$$

for a magnetic dipole $\vec{m}$. (magnet where we limit two points together).

We can generalize

$$A(\vec{x}) = \frac{\mu_0}{4\pi} \int_V d^3\vec{x}' \frac{\vec{M}(\vec{x}') \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} \quad (7)$$

we can use

$$\frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} = \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} \quad (8)$$

to get

$$= \frac{\mu_0}{4\pi} \int_V d^3\vec{x} \vec{M}(\vec{x}) \times \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} \quad (9)$$

IBP

$$A_k(\vec{x}) = \frac{\mu_0}{4\pi} \int_V d^3x' \sum_{ijk} \epsilon_{ijk} \vec{M}_i(\vec{x}') \partial'_j \frac{1}{|\vec{x} - \vec{x}'|} \quad (10)$$

$$= \text{surface term} - \frac{\mu_0}{4\pi} \sum_{ijk} \int_V \epsilon_{kij} \left(\partial'_j \vec{M}_i(\vec{x}') \right) \frac{1}{|\vec{x} - \vec{x}'|} \quad (11)$$

$$\approx -\frac{\mu_0}{4\pi} \int_V \sum_{ijk} (-\epsilon_{ijk}) \frac{\left(\partial'_j \vec{M}_i(\vec{x}') \right)}{|\vec{x} - \vec{x}'|} d^3\vec{x}' \quad (12)$$

$$= +\frac{\mu_0}{4\pi} \int \frac{\left(\vec{\nabla}' \times \vec{M} \right)_k(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3\vec{x}' \quad (13)$$

Where we have dropped the surface term and we have a potential of a current density (Biot savart law).

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int \frac{(\vec{\nabla}' \times \vec{M})_k(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3\vec{x}' \quad (14)$$

define

$$\vec{j}_{bound} = \vec{\nabla} \times \vec{M} \implies \vec{\nabla} \cdot \vec{j}_{bound} = 0 \quad (15)$$

$$\epsilon_{kij} \partial_j M_j = (\vec{\nabla} \times \vec{M})_k \quad (16)$$

$$\epsilon_{kij} \partial_k \partial_i M_j = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{M}) = 0 \quad (17)$$

vanishes as we contract antisymmetric with symmetric.

currents which satisfy this are called steady currents

Split

$$\vec{j} = \vec{j}_{free} + \vec{j}_{bound} = \vec{j}_{free} + \vec{\nabla} \times \vec{M} \quad (18)$$

Ampere's law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} - \mu_0 (\vec{j}_{free} + \vec{\nabla} \times \vec{M}) \quad (19)$$

Define $\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$ (magnetising/ auxiliary field). since cross products (rotations) are linear should be able to combine them.

$$\vec{\nabla} \times \vec{H} = \vec{j}_{free} \quad (20)$$

For para or diamagnetic materials we have

$$\vec{B} = \mu \vec{H} \quad (21)$$

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M} = \frac{\vec{B}}{\mu_0} \frac{\chi_m}{1 + \chi_m} \quad (22)$$

$$= \frac{1}{\mu_0 (1 + \chi_m)} \vec{B} \quad (23)$$

$$\implies \mu = \mu_0 (1 + \chi_m) = \mu_0 \mu_r \quad (24)$$

with $1 = \chi_m = \mu_r$.

$$\vec{H} = \frac{1}{\mu} \vec{B} = \frac{\mu_0}{\mu} (\vec{H} + \vec{M}) = \vec{M} = \left(\frac{\mu}{\mu_0} - 1 \right) \vec{H} = \chi_m \vec{H} \quad (25)$$

$$\vec{M} = \chi_m \vec{H} \quad (26)$$

So far we have talked about time independent systems.

For time dependent we will use current conservation $\vec{\nabla} \cdot \vec{j}_{bound} + \frac{\partial \rho_{bound}}{\partial t} = 0$
recall

$$\rho_{bound} = -\vec{\nabla} \cdot \vec{P} \quad (27)$$

$$\vec{\nabla} \cdot \vec{j}_{bound} - \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{P} = 0 \quad (28)$$

$$\implies \vec{j}_{bound} = \vec{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t} \quad (29)$$

$$(30)$$

Maxwell equation from Ampere's law

$$\vec{\nabla} \times \vec{B} \mu_0 - \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{j} = \mu_0 (\vec{j}_{free} + \vec{j}_{bound}) = \mu_0 \left(\vec{j}_{free} + \vec{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t} \right) \quad (31)$$

$$\vec{\nabla} \times (\vec{B} - \mu_0 \vec{M}) - \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\vec{E} + \frac{1}{\epsilon_0} \vec{P} \right) = \mu_0 \vec{j}_{free} \quad (32)$$

with $(\vec{B} - \mu_0 \vec{M}) = \mu_0 \vec{H}$ and $(\vec{E} + \frac{1}{\epsilon_0} \vec{P}) = \frac{1}{\epsilon_0} \vec{D}$.

$$\vec{\nabla} \times \vec{H} - \frac{\partial \vec{D}}{\partial t} = \vec{j}_{free} \quad (33)$$

remaining equations

$$\vec{\nabla} \cdot \vec{D} = \rho_{free} \quad (34)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (35)$$

$$\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 \quad (36)$$